

Exercise # 15.4

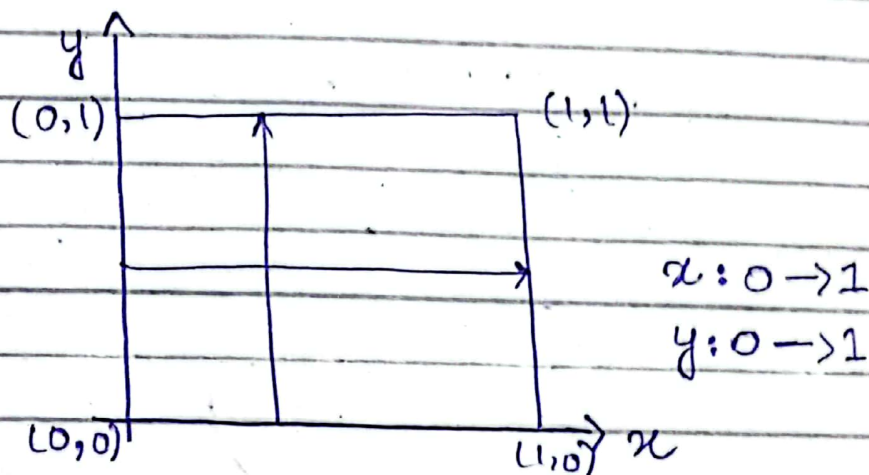
Green Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

Q No 1:- Evaluate line integral using Green's Theorem and check the answer by evaluating it directly.

$\oint y^2 dx + x^2 dy$, where C is the square with vertices $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$ oriented counter clockwise.

Sol)



Here,

$$f(x, y) = xy^2, \quad g(x, y) = x^2y$$
$$\frac{\partial f}{\partial y} = 2xy, \quad \frac{\partial g}{\partial x} = 2xy$$

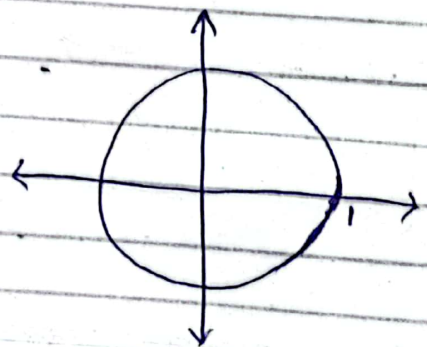
Applying Green's Theorem

$$\oint y^2 dx + x^2 dy = \iint_R (2y - 2x) dy dx$$
$$= \int_0^1 \int_0^1 (2y - 2x) dy dx$$

$$\begin{aligned}
&= \int_0^1 \left[\frac{2y^2}{2} - 2xy \right]_0^1 dx \\
&= \int_0^1 [y^2 - 2xy]_0^1 dx \\
&= \int_0^1 [1 - 2x - 0 + 0] dx \\
&= \int_0^1 [1 - 2x] dx \\
&= \left[x - \frac{2x^2}{2} \right]_0^1 \\
&= [1 - 1 - 0 + 0] \\
&= 0 \quad \text{Ans.}
\end{aligned}$$

Qno2) $\oint y dx + x dy$, where C is the unit circle oriented counter clockwise.

Sol)



$$\begin{aligned}
\frac{\partial f}{\partial y} &= \frac{y^2}{2} \\
\frac{\partial g}{\partial x} &= \frac{x^2}{2}
\end{aligned}$$

$r: 0 \rightarrow 1$ (unit circle)
 $\theta: 0 \rightarrow 2\pi$

Then,

$$\oint y dx + x dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

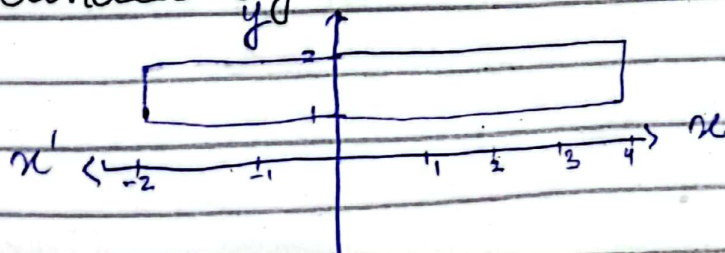
$$= \int_0^{2\pi} \int_0^1 \left(\frac{x^2}{2} - \frac{y^2}{2} \right) r dr d\theta$$

$$\begin{aligned}
&= \int_0^{2\pi} \int_0^1 \left[\frac{r^2 \cos^2 \theta}{2} - \frac{r^2 \sin^2 \theta}{2} \right] r dr d\theta \\
&= \int_0^{2\pi} \int_0^1 \left[\frac{r^3 \cos^2 \theta}{2} - \frac{r^3 \sin^2 \theta}{2} \right] dr d\theta \\
&= \int_0^{2\pi} \left[\frac{r^4 \cos^2 \theta}{8} - \frac{r^4 \sin^2 \theta}{8} \right]_0^1 d\theta \\
&= \int_0^{2\pi} \left[\frac{\cos^2 \theta}{8} - \frac{\sin^2 \theta}{8} \right] d\theta \\
&= \frac{1}{8} \int_0^{2\pi} \cos^2 \theta d\theta - \frac{1}{8} \int_0^{2\pi} \sin^2 \theta d\theta \\
&= \frac{1}{8} \int_0^{2\pi} \left[\frac{1 + \cos 2\theta}{2} \right] d\theta - \frac{1}{8} \int_0^{2\pi} \left[\frac{1 - \cos 2\theta}{2} \right] d\theta \\
&= \frac{1}{16} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi} - \frac{1}{16} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} \\
&= \frac{1}{16} \left[\frac{2\pi + \sin 4\theta}{2} - \frac{0 + \sin 0}{2} \right] - \frac{1}{16} \left[\frac{2\pi - \sin 4\theta}{2} - \frac{0 + \sin 0}{2} \right] \\
&= \frac{1}{16} \left[2\pi \right] - \frac{1}{16} \left[2\pi \right] \\
&= \frac{2\pi}{16} - \frac{2\pi}{16} = 0 \quad \text{Ans}
\end{aligned}$$

Use green's theorem to evaluate integral.
In each exercise, assume that the curve C is oriented counter clockwise.

Qno3) $\oint 3xy dx + 2xy dy$, where C is rectangle bounded by $x = -2$, $x = 4$, $y = 1$ and $y = 2$.

Sol)



Here,

$$f(x, y) = 3xy, \quad g(x, y) = 2xy$$

$$\frac{\partial f}{\partial y} = 3x, \quad \frac{\partial g}{\partial x} = 2y$$

Applying Green's theorem

$$\oint_C 3xy dx + 2xy dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$= \int_{-2}^2 \int_{-1}^2 (2y - 3x) dy dx$$

$$= \int_{-2}^2 \left[\frac{2y^2}{2} - 3xy \right]_{-1}^2 dx$$

$$= \int_{-2}^2 [y^2 - 3xy]_{-1}^2 dx$$

$$= \int_{-2}^2 [4 - 6x - 1 + 3x] dx$$

$$= \int_{-2}^2 [3 - 3x] dx$$

$$= \left[3x - \frac{3x^2}{2} \right]_{-2}^2$$

$$= \left[12 - \frac{48}{2} - \left[-6 - \frac{12}{2} \right] \right]$$

$$= \left[12 - \frac{48}{2} + 6 + \frac{12}{2} \right]$$

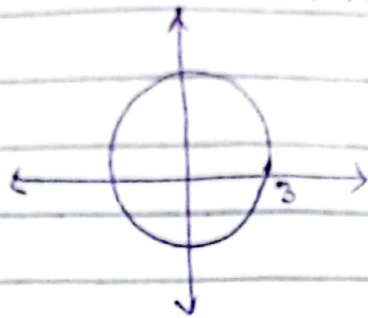
$$= 0 \text{ Ans:-}$$

4) $\oint_C (x^2 - y^2) dx + x dy$, where C is the circle $x^2 + y^2 = 9$.

Sol)

$$f(x, y) = x^2 - y^2 \quad , \quad g(x, y) = x$$

$$\frac{\partial f}{\partial y} = -2y \quad , \quad \frac{\partial g}{\partial x} = 1$$



Here,

$$x^2 + y^2 = 9$$

$$r^2 = 9$$

$$r = 3$$

$$r : 0 \rightarrow 3$$

$$\theta : 0 \rightarrow 2\pi$$

Applying Green's Theorem

$$\oint_C (x^2 - y^2) dx + x dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$= \int_0^{2\pi} \int_0^3 (1 + 2y) r dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 (1 + 2r \sin \theta) r dr d\theta$$

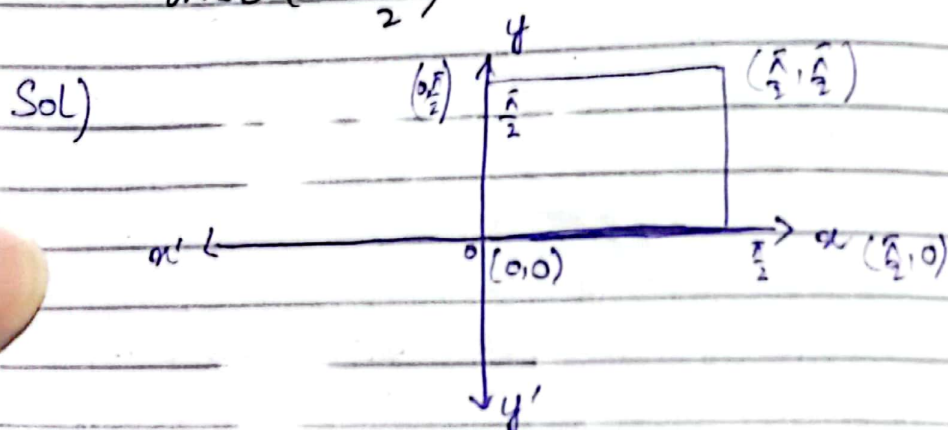
$$= \int_0^{2\pi} \int_0^3 (r + 2r^2 \sin \theta) dr d\theta$$

$$= \int_0^{2\pi} \left[\frac{r^2}{2} + \frac{2}{3} r^3 \sin \theta \right]_0^3 d\theta$$

$$= \int_0^{2\pi} \left[\frac{9}{2} + 18 \sin \theta \right] d\theta$$

$$\begin{aligned}
 &= \frac{9}{2} \int_0^{2\pi} d\theta + 18 \int_0^{2\pi} \sin\theta d\theta \\
 &= \frac{9}{2} [\theta]_0^{2\pi} + 18 [-\cos\theta]_0^{2\pi} \\
 &= \frac{9}{2} [2\pi] - 18 [\cos 2\pi - \cos 0] \\
 &= 9\pi \quad \text{Ans.}
 \end{aligned}$$

Q205) $\oint x \cos y dx - y \sin x dy$, where C is the square with vertices $(0,0)$, $(\frac{\pi}{2}, 0)$, $(\frac{\pi}{2}, \frac{\pi}{2})$ and $(0, \frac{\pi}{2})$



Here,

$$f(x,y) = x \cos y, \quad g(x,y) = -y \sin x$$

$$\frac{\partial f}{\partial y} = -x \sin y, \quad \frac{\partial g}{\partial x} = -y \cos x$$

Apply Green's Theorem

$$\oint_C x \cos y dx - y \sin x dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} (-y \cos x + x \sin y) dy dx$$

$$= \int_0^{\frac{\pi}{2}} \left[-\frac{y^2 \cos x}{2} - x \cos y \right]_{\frac{\pi}{2}}^{\frac{\pi}{2}} dx$$

$$= \int_0^{\frac{\pi}{2}} \left[-\frac{\pi^2 \cos x}{8} - x \left(\frac{\pi^2}{2} + 0 + x \right) \right] dx$$

$$= \int_0^{\frac{\pi}{2}} \left[-\frac{\pi^2 \cos x}{8} + x \right] dx$$

$$= \left[-\frac{\pi^2 \sin x}{8} + \frac{x^2}{2} \right]_0^{\frac{\pi}{2}}$$

$$= -\frac{\pi^2 \sin \frac{\pi}{2}}{8} + \frac{\pi^2}{8} + \frac{\pi^2 \sin 0}{8} - 0$$

$$= -\frac{\pi^2}{8} + \frac{\pi^2}{8} = 0$$

Ques) $\oint_C y \tan^2 x dx + \tan x dy$, where C

is the circle $x^2 + (y+1)^2 = 1$

Sol)

Here,

$$f(x, y) = y \tan^2 x, \quad g(x, y) = \tan x$$

$$\frac{\partial f}{\partial y} = \tan^2 x, \quad \frac{\partial g}{\partial x} = \sec^2 x$$

Green's theorem

$$= \iint_R (\sec^2 x - \tan^2 x) dA \quad \because 1 + \tan^2 x = \sec^2 x$$

$$= \iint_R dA \rightarrow \pi \text{ Ans.}$$

As

$$r = 1$$

$$A = \pi r^2$$

$$A = \pi$$

Q.7) $\oint_C (x^2 - y) dx + x dy$, where C is the circle $x^2 + y^2 = 4$.

Sol)

Here,

$$f(x, y) = x^2 - y, \quad g(x, y) = x$$

$$\frac{\partial f}{\partial y} = -1, \quad \frac{\partial g}{\partial x} = 1$$

As,

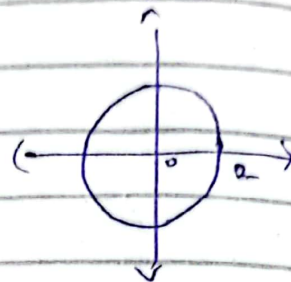
$$x^2 + y^2 = 4$$

$$r^2 = 4$$

$$r = 2$$

$$r: 0 \rightarrow 2$$

$$\theta: 0 \rightarrow 2\pi$$



Apply Green's theorem

$$\oint_C (x^2 - y) dx + x dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$= \int_0^{2\pi} \int_0^2 (1 + 1) r dr d\theta$$

$$= \int_0^{2\pi} \int_0^2 2r dr d\theta$$

$$= \int_0^{2\pi} \left[\frac{2r^2}{2} \right]_0^2 d\theta$$

$$= \int_0^{2\pi} [4] d\theta$$

$$= [4\theta]_0^{2\pi}$$

$$= 4(2\pi)$$

$$= 8\pi \text{ Ans:-}$$

Ques) $\oint_C (e^x + y^2) dx + (e^y + x^2) dy$, where C is the boundary on region $y = x^2, y = x$

(Sol)

$$f(x, y) = e^x + y^2, \quad g(x, y) = e^y + x^2$$

$$\frac{\partial f}{\partial y} = 2y, \quad \frac{\partial g}{\partial x} = 2x$$

Apply Green's Theorem

$x: 0 \rightarrow 1$
Remember

$$\oint_C (e^x + y^2) dx + (e^y + x^2) dy = \iint_D \left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x} \right) dA$$

$$= \int_0^1 \int_{x^2}^x (2x - 2y) dy dx$$

$$= \int_0^1 \left[2xy - \frac{2y^2}{2} \right]_{x^2}^x dx$$

$$= \int_0^1 \left[2xy - y^2 \right]_{x^2}^x dx$$

$$= \int_0^1 [2x^2 - x^2 - 2x^3 + x^4] dx$$

$$= \int_0^1 [x^2 - 2x^3 + x^4] dx$$

$$= \left[\frac{x^3}{3} - \frac{2x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= \left[\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5} \right]_0^1$$

$$= \left[\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right]$$

$$= \frac{1}{30} \text{ Ans!}$$

Equation of line AB is $\frac{x-1}{2-1} = \frac{y-0}{0-0}$ which

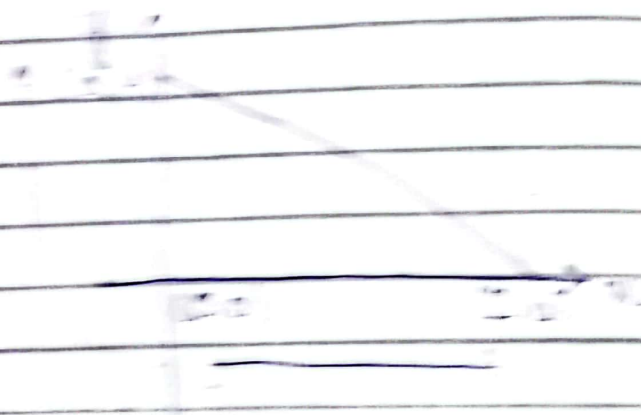
is the straight line $y=0$ vertices
 $(1,0)$, $(2,0)$ and $(0,0)$

Sol

Let

$$\frac{x-1}{2-1} = \frac{y-0}{0-0} \Rightarrow \frac{x-1}{1} = \frac{y}{0} = -\frac{1}{0}$$

$$\frac{x-1}{1} = \frac{y}{0} = -\frac{1}{0}$$



$$m = \frac{0-0}{2-1} = \frac{0-0}{2-1} = \frac{0}{1} = 0$$

$$y-0 = m(x-1)$$

$$y-0 = 0(x-1)$$

$$y = 0$$

$$x = 1-2$$

$$x = 0-20$$

Rem.

$$\int_C \ln(1+y) dx - \frac{xy}{(1+y)} dy \cdot \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

$$= \int_0^2 \int_0^{2x} \left[\frac{-y}{1+y} - \frac{1}{1+y} \right] dy dx$$

$$\frac{-1}{1+y} \sqrt{-y}$$

$$\frac{-1}{1+y} + \frac{1}{1+y}$$

$$= \int_0^2 \left[\int_0^{2x} -1 + \frac{1}{1+y} \right] dy - \left[\int_0^{2x} \frac{1}{1+y} \right] dy$$

$$= \int_0^2 \left[-y + \ln(1+y) \right]_0^{2x} - \left[\ln(1+y) \right]_0^{2x}$$

$$= \int_0^2 -2x dx + \int_0^2 \ln(1+y) \Big|_0^{2x} dx - \int_0^2 \ln(1+y) \Big|_0^{2x} dx$$

$$= -\frac{2x^2}{2} \Big|_0^2$$

$$= -4 \text{ ans:-}$$

Qno 10) $\oint_C x^2 y dx - y^2 x dy$, where C is the boundary of the region in the 1st quadrant, enclosed by the co-ordinate axes and the circle $x^2 + y^2 = 16$.

Sol)

$$f(x, y) = x^2 y, \quad g(x, y) = -y^2 x$$

$$\frac{\partial f}{\partial y} = x^2, \quad \frac{\partial g}{\partial x} = -y^2$$

Here,

$$x^2 + y^2 = 16$$

$$r^2 = 16$$

$$r = 4$$

$$r: 0 \rightarrow 4$$

Due to 1st Quad

$$\theta: 0 \rightarrow \frac{\pi}{2}$$

Apply Green's Theorem

$$\oint_C x^2 y dx - y^2 x dy = \iint_R \left(\frac{\partial f}{\partial y} - \frac{\partial g}{\partial x} \right) dA$$

$$= \frac{\pi}{2} \int_0^4 \int_0^{\pi/2} (-y^2 - x^2) r dr d\theta$$

$$= -\frac{\pi}{2} \int_0^4 \int_0^{\pi/2} (y^2 + x^2) r dr d\theta$$

$$= -\frac{\pi}{2} \int_0^4 \int_0^{\pi/2} (r^2 \sin^2 \theta + r^2 \cos^2 \theta) r dr d\theta$$

$$= -\frac{\pi}{2} \int_0^{\pi} \int_0^4 r^2 (\sin^2 \theta + \cos^2 \theta) r \, dr \, d\theta$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$= -\int_0^{\pi} \int_0^4 r^3 \, dr \, d\theta$$

$$= -\frac{\pi}{2} \left[\frac{r^4}{4} \right]_0^4 d\theta$$

$$= -\frac{\pi}{2} \int_0^{\pi} 64 \, d\theta$$

$$= -64 \int_0^{\pi} d\theta$$

$$= -64 [\theta]_0^{\pi}$$

$$= -64 \left[\frac{\pi}{2} \right]$$

$$= -32\pi \text{ Ans.}$$